Big Data Analytics
CISELEC 501A-CS41S1

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09/13/2025

Lab Activity-03: CRISP-DM | Exploring CRISP-DM Phases 1, 2, & 3 using Bank Marketing Data (Total Points: 65)

OBJECTIVE: This activity is designed to explore the initial phases of the CRISP-DM framework – Business Understanding, Data, Understanding, and Data Preparation – through scenario-based analysis. It involves identifying real-world problems in the banking sector, examining a publicly available dataset, and preparing the data for future modeling.
The exercise emphasizes the importance of aligning data mining efforts with clearly defined business goals, resource considerations, and contextual constraints, forming a foundation for informed decision-making and effective analytical planning.

CRISP-DM: Step-2 | Data Understanding

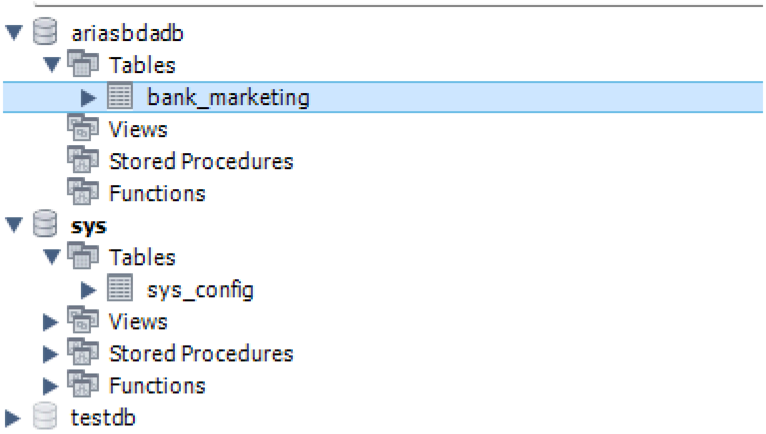
Instructions:

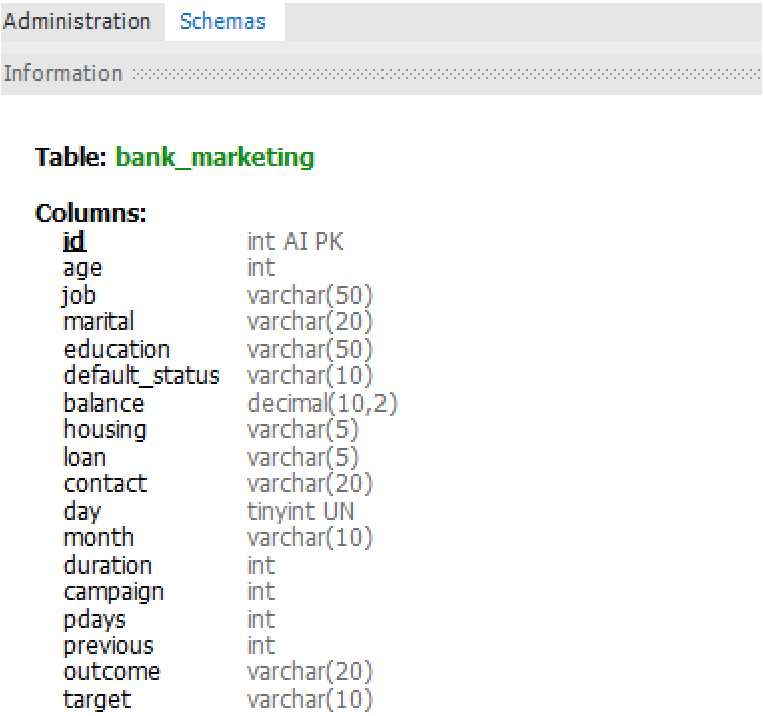
1. Download the MySQL data-dump located at
   https://github.com/AI-Artisans/bda_cs41s1/tree/main/20250913-Lab-Wk8

2. From the command line (cmd.exe), restore the MySQL data-dump using the command below:
   mysql -u [MySQL DB User] [MySQL database] < [local path of MySQL data-dump file

   example:
   mysql -u root testdb < D:\20250913-Laboratory.sql

3. After the data-dump restoration, explore the newly created table and its contents.


ariasbdadb
Tables
bank_marketing
Views
Stored Procedures
Functions
sys
Tables
sys_config
Views
Stored Procedures
Functions
testdb


Administration
Schemas
Information
Table: bank_marketing
Columns:
id int AI PK
age int
job varchar(50)
marital varchar(20)
education varchar(50)
default_status varchar(10)
balance decimal(10,2)
housing varchar(5)
loan varchar(5)
contact varchar(20)
day tinyint UN
month varchar(10)
duration int
campaign int
pdays int
previous int
outcome varchar(20)
target varchar(10)

```
Command Prompt - "C:\Progr    Command Prompt - mysqld    Command Prompt
mysql> USE ariasbdadb;
Database changed
mysql> SHOW TABLES;
+----------------------+
| Tables_in_ariasbdadb |
+----------------------+
| bank_marketing       |
+----------------------+
1 row in set (0.00 sec)

mysql> DESCRIBE bank_marketing;
+----------------+------------------+------+-----+---------+----------------+
| Field          | Type             | Null | Key | Default | Extra          |
+----------------+------------------+------+-----+---------+----------------+
| id             | int              | NO   | PRI | NULL    | auto_increment |
| age            | int              | NO   |     | NULL    |                |
| job            | varchar(50)      | YES  |     | NULL    |                |
| marital        | varchar(20)      | YES  |     | NULL    |                |
| education      | varchar(50)      | YES  |     | NULL    |                |
| default_status | varchar(10)      | YES  |     | NULL    |                |
| balance        | decimal(10,2)    | YES  |     | NULL    |                |
| housing        | varchar(5)       | YES  |     | NULL    |                |
| loan           | varchar(5)       | YES  |     | NULL    |                |
| contact        | varchar(20)      | YES  |     | NULL    |                |
| day            | tinyint unsigned | YES  |     | NULL    |                |
| month          | varchar(10)      | YES  |     | NULL    |                |
| duration       | int              | YES  |     | NULL    |                |
| campaign       | int              | YES  |     | NULL    |                |
| pdays          | int              | YES  |     | NULL    |                |
| previous       | int              | YES  |     | NULL    |                |
| outcome        | varchar(20)      | YES  |     | NULL    |                |
| target         | varchar(10)      | YES  |     | NULL    |                |
+----------------+------------------+------+-----+---------+----------------+
18 rows in set (0.00 sec)

mysql> SELECT * FROM bank_marketing LIMIT 10;
+----+-----+--------------+----------+-----------+----------------+---------+---------+------+---------+-----+-------+----------
+----------+-------+----------+---------+--------+
| id | age | job          | marital  | education | default_status | balance | housing | loan | contact | day | month | duration
| campaign | pdays | previous | outcome | target |
+----+-----+--------------+----------+-----------+----------------+---------+---------+------+---------+-----+-------+----------
+----------+-------+----------+---------+--------+
```

```
Command Prompt - "C:\Progr    Command Prompt - mysqld    Command Prompt
18 rows in set (0.00 sec)

mysql> SELECT * FROM bank_marketing LIMIT 10;
+----+-----+--------------+----------+-----------+----------------+---------+---------+------+---------+-----+-------+----------
+----------+-------+----------+---------+--------+
| id | age | job          | marital  | education | default_status | balance | housing | loan | contact | day | month | duration
| campaign | pdays | previous | outcome | target |
+----+-----+--------------+----------+-----------+----------------+---------+---------+------+---------+-----+-------+----------
+----------+-------+----------+---------+--------+
|  1 |  58 | management   | married  | tertiary  | no             | 2143.00 | yes     | no   | unknown |   5 | may   |      261
|        1 |    -1 |        0 | unknown | no     |
|  2 |  44 | technician   | single   | secondary | no             |   29.00 | yes     | no   | unknown |   5 | may   |      151
|        1 |    -1 |        0 | unknown | no     |
|  3 |  33 | entrepreneur | married  | secondary | no             |    2.00 | yes     | yes  | unknown |   5 | may   |       76
|        1 |    -1 |        0 | unknown | no     |
|  4 |  47 | blue-collar  | married  | unknown   | no             | 1506.00 | yes     | no   | unknown |   5 | may   |       92
|        1 |    -1 |        0 | unknown | no     |
|  5 |  33 | unknown      | single   | unknown   | no             |    1.00 | no      | no   | unknown |   5 | may   |      198
|        1 |    -1 |        0 | unknown | no     |
|  6 |  35 | management   | married  | tertiary  | no             |  231.00 | yes     | no   | unknown |   5 | may   |      139
|        1 |    -1 |        0 | unknown | no     |
|  7 |  28 | management   | single   | tertiary  | no             |  447.00 | yes     | yes  | unknown |   5 | may   |      217
|        1 |    -1 |        0 | unknown | no     |
|  8 |  42 | entrepreneur | divorced | tertiary  | yes            |    2.00 | yes     | no   | unknown |   5 | may   |      380
|        1 |    -1 |        0 | unknown | no     |
|  9 |  58 | retired      | married  | primary   | no             |  121.00 | yes     | no   | unknown |   5 | may   |       50
|        1 |    -1 |        0 | unknown | no     |
| 10 |  43 | technician   | single   | secondary | no             |  593.00 | yes     | no   | unknown |   5 | may   |       55
|        1 |    -1 |        0 | unknown | no     |
+----+-----+--------------+----------+-----------+----------------+---------+---------+------+---------+-----+-------+----------
+----------+-------+----------+---------+--------+
10 rows in set (0.00 sec)
       SHOW TABLES;
mysql>
```

REFLECTION:

1. **What challenges did you experience during the MySQL data-dump restoration? (5 pts)**
    - One of the challenges I encountered during data dump restoration is the fact that i am using a guest pc. Of course, in a guest pc, I won't have the rights to setup a server in which mysql needs to have. Unlike sqlite where it is serverless, I have to establish a server but I don't have admin rights to do so.

      Luckily, in mysql it has an ODBC connection. Since mysql is already installed and has its driver for the ODBC, I can use this to set up my connection locally.

2. **With respect to the newly created MySQL table, complete the details below (5 pts):**
    -

Number of Attributes	18
Number of Objects	47277

-

3. **With respect to the objects inside the table, complete the details below (10 pts):**
    -

Attribute	**Data Type**
id	int
age	int
job	varchar (50)
marital	varchar (20)
education	varchar (50)
default_status	varchar (10)
balance	decimal(10,2)
housing	varchar (5)
loan	varchar (5)
contact	varchar (20)
day	tinyint unsigned
month	varchar (10)
duration	int
campaign	int
pdays	int
previous	int

outcome	varchar (20)
target	varchar (10)

-

CRISP-DM: Step-3 | Data Preparation

Instructions:

1. **Perform initial data exploration using the restored data-dump. Complete item #1 in Reflection section**
2. **Based on the summary produced in Reflection item #1, kindly perform data clean-up on the following attributes:**
   - Job
   - Marital
   - Education
   - Contact
3. **Perform another set of data exploration after the data clean-up and complete item #2 in Reflection section.**

```
mysql> select job, count(*)
+---------------+------+
| job           | ctr  |
+---------------+------+
| management    | 9458 |
| technician    | 7597 |
| entrepreneur  | 1487 |
| blue-collar   | 9732 |
| unknown       |  288 |
| retired       | 2264 |
| admin         | 5171 |
| services      | 4154 |
| self-employed | 1579 |
| unemployed    | 1303 |
| housemaid     | 1240 |
| student       |  938 |
+---------------+------+
```

```
mysql> SELECT marital,
+----------+-------+
| marital  | count |
+----------+-------+
| married  | 27214 |
| single   | 12790 |
| divorced |  5207 |
+----------+-------+
```

```
mysql> SELECT education,
+-----------+-------+
| education | count |
+-----------+-------+
| tertiary  | 13315 |
| secondary | 23202 |
| unknown   |  1843 |
| primary   |  6851 |
+-----------+-------+
4 rows in set (0.02 sec)
```

```
mysql> SELECT Contact, C
+-----------+-------+
| Contact   | count |
+-----------+-------+
| unknown   | 13020 |
| cellular  | 29285 |
| telephone |  2906 |
+-----------+-------+
3 rows in set (0.02 sec)
```

REFLECTION:

1. **Provide the initial summary for the following attributes (10 pts):**
    - Job

```
+---------------+------+
| job           | ctr  |
+---------------+------+
| management    | 9458 |
| technician    | 7597 |
| entrepreneur  | 1487 |
| blue-collar   | 9732 |
| unknown       |  277 |
| retired       | 2170 |
| admin         | 5171 |
| services      | 3639 |
| self-employed | 1579 |
| unemployed    | 1303 |
| housemaid     | 1240 |
| student       |  938 |
| svcs          |  515 |
| Ret           |   94 |
| ???           |   11 |
+---------------+------+
```

    - Marital

marital	count
married	27214
single	12790
divorced	4624
Div	583

- Education

education	count
tertiary	13272
secondary	23082
unknown	1843
primary	6851
2nd	120
3rd	29
???	14

- Contact

Contact	count
unknown	13020
cellular	28947
telephone	2719
Tel	187
cell.	338

2. **Provide the final summary for the following attributes (15 pts) :**
   - Job

```
mysql> select job, count(*)
+---------------+------+
| job           | ctr  |
+---------------+------+
| management    | 9458 |
| technician    | 7597 |
| entrepreneur  | 1487 |
| blue-collar   | 9732 |
| unknown       |  288 |
| retired       | 2264 |
| admin         | 5171 |
| services      | 4154 |
| self-employed | 1579 |
| unemployed    | 1303 |
| housemaid     | 1240 |
| student       |  938 |
+---------------+------+
```

- Marital

```
mysql> SELECT marital,
+----------+-------+
| marital  | count |
+----------+-------+
| married  | 27214 |
| single   | 12790 |
| divorced |  5207 |
+----------+-------+
```

- Education

```
mysql> SELECT education,
+-----------+-------+
| education | count |
+-----------+-------+
| tertiary  | 13315 |
| secondary | 23202 |
| unknown   |  1843 |
| primary   |  6851 |
+-----------+-------+
4 rows in set (0.02 sec)
```

- Contact

```
mysql> SELECT Contact, C
+-----------+-------+
| Contact   | count |
+-----------+-------+
| unknown   | 13020 |
| cellular  | 29285 |
| telephone |  2906 |
+-----------+-------+
3 rows in set (0.02 sec)
```

3. **Based on the results produced in items #1 and #2, what issues did you encounter during the clean-up process? (5 pts)**

   - One of the issues I encountered during the clean-up is how do I actually clean those objects in their respective attributes. My professor helped me visualize this by exploring the counts of each value of an object that appeared in their respective attributes.

4. **What are the possible repercussions if data clean-up is neglected during the "Data Preparation" phase of the CRISP-DM framework? (10 pts)**

   - Possible repercussions will be the skewers or skewed counts. During that phase, we see that there are categories that are same but different name or are somehow abbreviated. If we neglect this, we will have skews count as it will be treated as different categories. We will have an inaccurate analysis of our datasets.

NOTE: Kindly upload all your work (excel and latest MySQL db-dump) to our GitLab repository following the

branch structure below (5 pts):

[Your GitLab Branch]

|- Lab-20250806

|- Lab-20250913

        |- [your excel file]

        |- [your latest db-dump]

|- README.md

" I affirm that I have not given or received any unauthorized help in this assignment, and that this work is my own "

Kenneth Arias